Roll No. 406ET031

Course: B.Tech. Computer Engineering/
B. Tech. Computer Science and Engineering (AI/ML)
Subject: Engineering Mathematics-II,
Subject Code: ETMT-102
Semester: II

Time: 03 Hours **Max Marks: 70**

Instructions to the Students:

1. This Question paper consists of two Sections. All sections are compulsory.
2. Section A comprises 10 questions of short answer type. All questions are compulsory. Each question carries 02 marks.
3. Section B comprises 8 long answer type questions out of which students must attempt any 5. Each question carries 10 marks.
4. Do not write anything on the question paper.

Q.No.	SECTION –A (SHORT ANSWER TYPE QUESTIONS)	Marks
1. a.	Evaluate $\int_0^1 \int_0^1 xy\, dx\, dy$.	(2)
b.	Evaluate $\iint_R (x+y)dA$, where dA indicates the small change in area in xy plane and R is the region bounded by the lines $x = 0, x = 2, y = x$ and $y = x + 2$.	(2)
c.	Solve $\sin x \frac{dy}{dx} + 2y = \left(\tan\frac{x}{2}\right)^3$.	(2)
d.	Solve $(1 + \log xy)dx + \left(1 + \frac{x}{y}\right)dy = 0$.	(2)
e.	Form a partial differential equation from $x^2 + y^2 + (z-c)^2 = a^2$.	(2)
f.	Solve $\frac{\partial^2 z}{\partial x \partial y} = x^2 y$ subject to the condition $z(x,0) = x^2$ and $z(1,y) = \cos y$.	(2)
g.	Find the Laplace transform of $(\cos t)^2$.	(2)
h.	Find the inverse Laplace transform of $\frac{1}{2s(s-3)}$.	(2)
i.	If $z = x^2y + 3xy^4$, where $x = \sin 2t$ and $y = \cos t$. Find $\frac{dz}{dt}$ when $t = 0$.	(2)
j.	If $f(x,y) = \frac{xy}{x^2+y^2}$. Does limit $\lim_{(x,y)\to(0,0)} f(x,y)$ exists?	(2)

SECTION –B (LONG ANSWER TYPE QUESTIONS)

2. a) Transform the integrals into cartesian form and hence evaluate $\int_0^{\pi}\int_0^{a} r^3 \sin\theta\cos\theta \; dr\, d\theta$. (5)

b) Evaluate $\int_0^{a}\int_y^{a} \frac{x}{x^2+y^2}dxdy$ by changing the order of integration (5)

3. a) Solve $(y^4+2y)dx + (xy^3+2y^4-4x)dy = 0$. (5)

b) Solve $\frac{d^2y}{dx^2} + 6\frac{dy}{dx} + 9y = \frac{e^{-3x}}{x^3}$. (5)

4. Find the extreme values of the function $f(x,y) = x^2+2y^2$ on the circle $x^2+y^2=1$. (10)

5. Find the centre of gravity of the area in the positive quadrant of the curve $x^{\frac{2}{3}} + y^{\frac{2}{3}} = a^{\frac{2}{3}}$. (10)

6. Solve the equation (10)

$$\frac{\partial^3 z}{\partial x^3} - 7\frac{\partial^3 z}{\partial x \partial y^2} - 6\frac{\partial^3 z}{\partial y^3} = \sin(x+2y) + e^{2x+y}.$$

7. a) Solve $(p^2+q^2)y = qz$ using Charpit's method. (5)

b) Find the Laplace transform of $\frac{1-\cos t}{t^2}$. (5)

8. Solve the initial value problem (10)

$2y'' + 5y' + 2y = e^{-2t}, y(0) = 1$ and $y'(0) = 1$ using Laplace transforms.

9. Solve by the method of variation of parameters: (10)

$$\frac{d^2y}{dx^2} - y = \frac{1}{1+e^x}.$$

===END OF PAPER===